Choosing the right tools

AI & Research, Evening 1

Gjorgji Strezoski Marleen de Jonge

Faculty of Economics and Business (FEB), University of Amsterdam

21 October 2026



EDITORIAL | 25 March 2026

AI scientists are changing research – institutions, funders and publishers must respond

The ability to automate the discovery process in some areas of scientific inquiry raises unanswered questions about how research should be conducted.

COMMENT | 29 September 2026

AI can widen science – but only if institutions stop rewarding the already measurable

Funders, journals and universities must reward the creation of new scientific terrain, not just the efficient mining of old ground.

OPINION | EVOLUTION

AI will reorganize science. Will research remain a human enterprise?

Michael F. Hochberg and Peter H. Thrall Authors Info & Affiliations

July 29, 2026 | 123 (31) e2610088123 | https://doi.org/10.1073/pnas.2610088123

ARTIFICIAL INTELLIGENCE

AI professors are negotiating the new realities of academic research

At a convening for the Schmidt Sciences AI2050 program, I saw how academic researchers are facing up to the challenges of the AI era.

By Grace Huckins August 10, 2026

SHORT WAVE | SCIENCE | LISTEN & FOLLOW

AI is helping individual scientists, study suggests — but not science

FEBRUARY 17, 2026 · 4:42 PM ET

HEARD ON ALL THINGS CONSIDERED

By Katia Riddle

Leaders | EurekAI!

What if the machines take over science?

Artificial intelligence will require scholars to reinvent their disciplines

Illustration: *The Economist*, Leaders


NEW IDEA!
MORE DATA
DIFFERENT METHOD?
WRITE INTRO
ANALYZE RESULTS
REVISE AGAIN
RESULTS (?)
PLAN
More data
Try this method
Revise draft
Check ref's
You got this!
PAPERS
LITERATURE REVIEW
IDEAS
DATA
ANALYSIS
DRAFTS
CODING
DATA CLEANING
LITERATURE SEARCH
WRITING
PLOTTING
REVIEWING

Aim of the workshop

- Research work is changing, whether we asked for it or not
- That is no reason to transform everything you do at once; conventional skills remain relevant
- Tonight's goal: know which tools exist, and what each one promises and risks, so you can make your work easier/faster/more fun
- We get there through one well-contained task: turn a paper into a newspaper article
  - Small on purpose, so we can try many options without getting lost in the chaos
  - From a single chat window up to agents working in your own folders
  - Demo first, then you do it yourself
- Did everyone select a paper for the assignment?



Materials

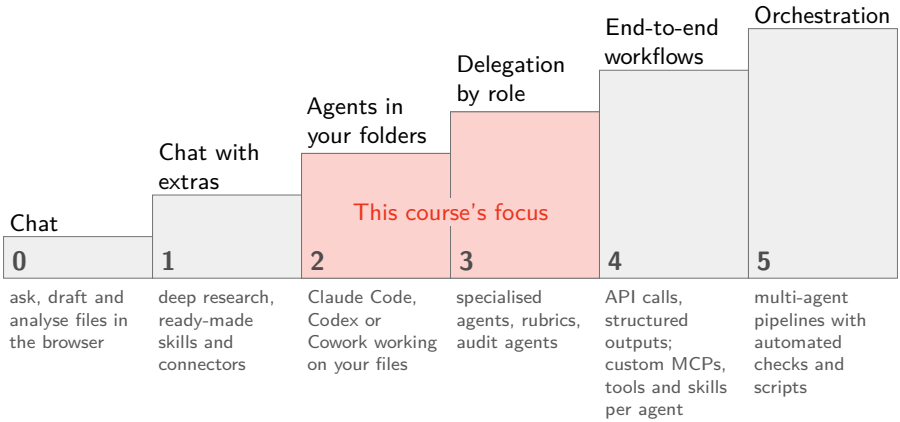
Example FD and FT articles, skill and agent files. Log in with your UvA account.

Materials for tonight

Paper, example FD and FT articles, prompts and the fd-article skill

Log in with your UvA account

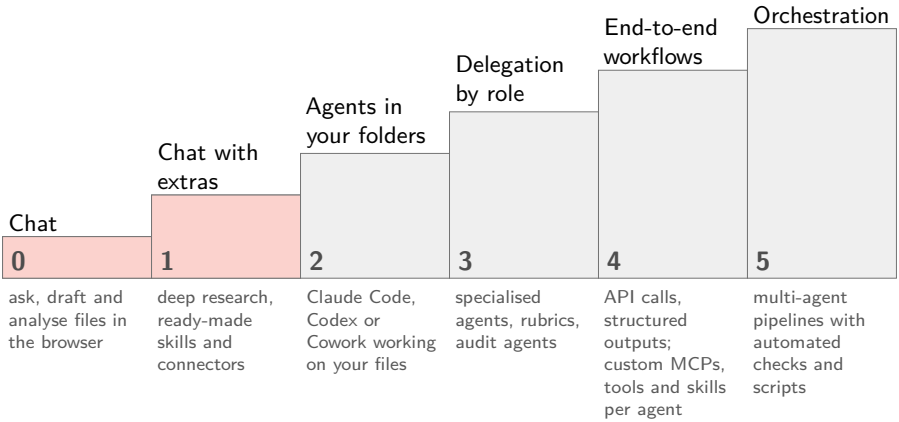
Using AI tools can be as complicated as you like



Assignment: turn a paper into an FD or FT article

- Write a newspaper article on one research paper, in the style of the Financieele Dagblad (FD) or the Financial Times (FT)
  - Your own paper, or one you find online
- Why this skill: you will often need to translate nuanced, complicated research into plain language
  - The same text has to work for experts, policy makers, and the general public
- How we work on it tonight: several ways to use AI for the same task, one tool at a time
  - From a single chat prompt up to agents executing standardized workflows in your own folders
  - Demo first, then you do it yourself

The familiar way first: a chat window



Four ways to let an AI work on your files

Where	Examples	Good for
Desktop app	Claude Code or Cowork in the Claude app	Starting out: pick a folder, no terminal
Code editor	VS Code with Claude Code, Codex or Copilot	Code, data and LaTeX: see every change the agent makes
Terminal	Claude Code, Codex CLI, Gemini CLI	Most control; automating repeated jobs
Browser	claude.ai/code, Codex in ChatGPT	Nothing to install; works on a copy in the cloud

Same agent underneath; the difference is how much you see and control.

Step 1: write the article with one prompt

paper.pdf | fd-example-1.pdf | fd-example-2.pdf | fd-example-3.pdf

You are an experienced science journalist at the Financieele Dagblad (FD).

Based on the paper, write a news article of about 600 words for the FD.

- *Use the three example articles as the benchmark for tone, structure and the way research is explained. Do not take any content from them.*
- *The piece must be easy to follow for a broad audience, but also hold up when an expert reads it.*
- *Find the policy relevance and open with that core message: the consequences for policy, the real economy and/or the financial sector.*
- *Use only findings and figures from the paper. Do not invent numbers or quotes.*

. . .

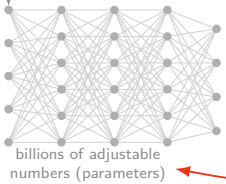
Sonnet 5.5 · effort: high ∨

At its core, a chat model learns to predict the next word

1. Hide the next word

The central bank raised interest [rates]

2. A network of parameters scores every word



rates	0.48
again	0.21
sharply	0.14
today	0.09

3. Compare with the real word

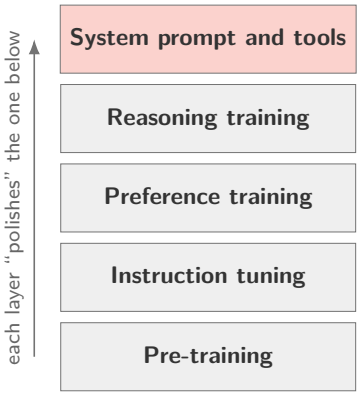
Gave "rates" only 0.48:
error = its distance from 1

4. Adjust the parameters

Nudge billions of numbers
so "rates" gets a little more
likely next time

- Repeat for trillions of words: grammar, facts and style emerge as by-products

Today's AI chats use much more than next-word guessing



The product around the model: hidden instructions, web search, reading your PDF, running code

Rewarded for getting checkable problems (maths, code) right: learns to work step by step

People and AI judges rate answers: learns to be helpful, honest, and to refuse some requests

Trained on example conversations: answers your question instead of continuing your text

Predict the next word on much of the internet. Gives language, knowledge and **plausible** text, not true text

Back to the demo: what did one prompt produce?

Banks — Financial stability · Research

Credit risk

Supply chain shocks could multiply bank losses fivefold, study finds

The collapse of one key supplier could wipe out a fifth of banks' capital, research using data on almost every company in Hungary shows

Science correspondent — Journal of Financial Stability study

Banks may be badly underestimating how much they could lose when a crisis in the real economy spreads through corporate supply chains, according to research that traced almost every trading link between companies in an entire country.

Once the knock-on effects of failing suppliers and customers are counted, banks' expected losses on their corporate loan books are on average 5.2 times higher than standard estimates, the study in the Journal of Financial Stability found.

The researchers, from the Complexity Science Hub Vienna, WU Vienna and the Central Bank of Hungary, said current practice "could systematically and grossly" underestimate the capital banks need and the interest rates they should charge.

Banks typically judge a borrower by its own balance sheet. But a sound company can still default if a key supplier stops delivering or a big customer stops buying, and that shock can travel far beyond a borrower's direct trading partners.

Back to the demo: what did one prompt produce?

Supply chain shocks could multiply bank losses fivefold, study finds (continued)

To measure the effect, the authors combined Hungarian tax records of the 2019 trades of 243,339 firms with the national credit register and the capital of 27 banks holding 84 per cent of corporate loans.

About 100 of those firms turned out to be systemically risky. The failure of any one of the 24 riskiest could wipe out between 10 and 22 per cent of the banking system's core capital, assuming lenders recover nothing from defaulted loans.

Almost all of those losses come not from the failing firm's own debts but from the defaults it sets off among other companies. Most of the riskiest firms are in energy, telecoms and transport.

Many of them borrow little or nothing from Hungarian banks, while the biggest borrowers tend to cause only small knock-on losses. "Firms with the largest bank-loans might not be the most relevant for monitoring financial stability," the authors wrote.

Supply chain effects make bad outcomes much worse. Across 10,000 simulated scenarios, banks' value at risk, the loss exceeded only in the worst 5 per cent of cases, rose on average by a factor of 6.7.

Most of the damage comes from companies that remain solvent but run out of cash to pay their short-term bills. They account for about 20 percentage points of the worst-case loss of 22 per cent.

Back to the demo: what did one prompt produce?



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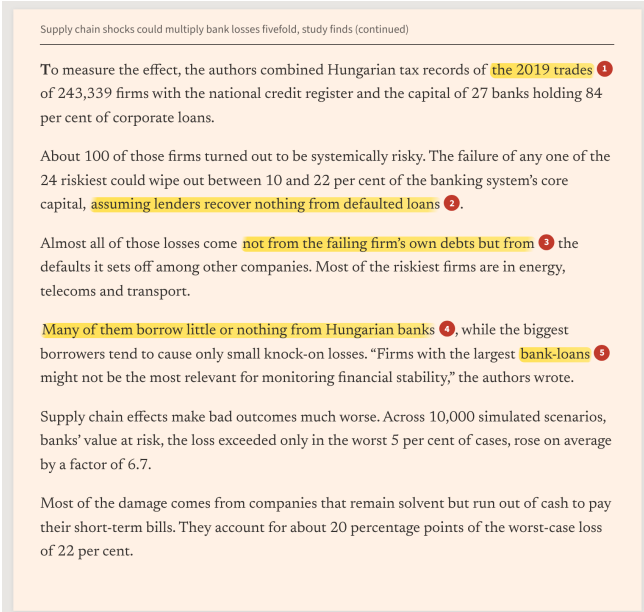
Once the knock-on effects of failing suppliers and customers are counted, banks' expected losses on their corporate loan books are on average 5.2 times higher than standard estimates [3], the study in the Journal of Financial Stability found [4].

The researchers, from the Complexity Science Hub Vienna, WU Vienna and the Central Bank of Hungary [5], said current practice "could systematically and grossly" [6] underestimate the capital banks need and the interest rates they should charge.

Banks typically judge a borrower by its own balance sheet. But a sound company can still default if a key supplier stops delivering or a big customer stops buying [7], and that shock can travel far beyond a borrower's direct trading partners.

1. **Vague but also dramatic**
   "May be" plays it safe, "badly" adds drama.
2. **Hides a key fact**
   Which country? Does this apply to us?
3. **Overclaim**
   The paper compares its model with and without contagion, not with banks' own models.
4. **No news hook**
   Published in 2024. Why is this news now?
5. **No people**
   Institutions, but no named researcher and no interview.
6. **Quote copied from the paper**
   Nobody was interviewed. No outside expert either.
7. **Abstract, not concrete**
   The paper mentions Katrina, Japan 2011 and Covid. None used.

Back to the demo: what did one prompt produce?



Supply chain shocks could multiply bank losses fivefold, study finds (continued)

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Most of the damage comes from companies that remain solvent but run out of cash to pay their short-term bills. They account for about 20 percentage points of the worst-case loss of 22 per cent.

1 Dated data, unflagged
Pre-Covid, pre-energy crisis. Still valid today? Not asked.

2 Big caveat in a subclause
The 22% is a worst case. A journalist would ask how realistic it is.

3 Typical LLM phrasing
A pattern LLMs overuse: "not X but Y".

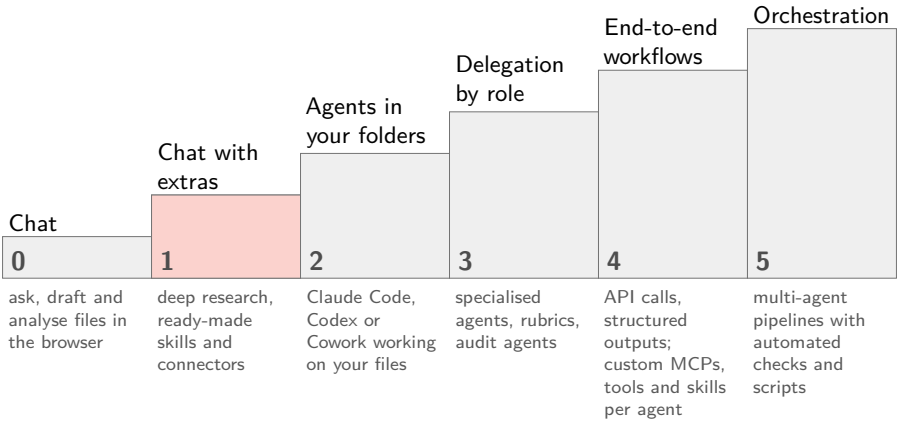
4 Best finding buried
The most surprising result sits in paragraph 8. A journalist would lead with it.

5 Paper's wording copied word for word
"bank-loans" is how the paper writes it, not how a newspaper would.

Impressive! So is that it for today?

- No
- There is still a lot to improve
  - We offloaded all the cognitive effort: the model did the thinking, not us
  - Nothing has been fact-checked: numbers, claims and sources are unverified
  - No human creativity: no voice, no angle, no story only we could tell
  - No reason to care: why this matters to a reader *now* is missing
  - Not yet ours to publish: we could not defend every sentence in it

One level up: the same chat, with extra tools



What a skill is

A skill is a **reusable instruction file** for a task you repeat.

Instead of re-explaining the task every time, you write down once:

- what the task is, and when it applies
- which steps to follow
- what the output should look like
- which checks and limits matter

Possible skills for tonight's assignment:

- an interview: ask you what the finding is and what message it must carry
- a news scan: find recent news on the topic, so you do not miss something obviously relevant
- a skeleton: one line per paragraph, for you to edit before any writing
- a house style: the structure, length and headline rules of the FD or FT
- a fact-check: trace every number to a page of the paper

Structure adapted from Paul Goldsmith-Pinkham, "Skills: Claude Code for Economists", Markus Academy, Ep. 162-6.

What a skill looks like

```
# SKILL.md
name: fd-article
description: turn one academic paper
into an FD- or FT-style news article.
Use when a researcher asks for a news
article about their paper.

## Steps
1. Read the paper. Show the author a
claim card: finding, key number, who
should care, what the paper does not
show.
2. Interview the author: five
questions. The answers are the only
source of quotes.
3. Draft to the house style (below).
4. Check: every number to a page, every
quote to an answer.
5. Revise on the author's corrections.

## Style (FD)
Headline is a claim. Three-bullet "In
het kort". Finding and institution in
the first sentence. Subhead every 2-3
paragraphs. One caveat paragraph.
500-700 words.

## Never
Invent a number, a quote or a person.
```

- A plain text file: when to use it, the steps, the style rules
- Loaded only when your request matches, so it does not clutter every chat
- You write it once, use it on every paper, and hand it to a colleague
- The prompt from step 1, with the lessons from the result folded in: it interviews you first and checks itself after
- To run it: type `/fd-article`, or name it in your request: *"Use the fd-article skill to turn my paper into an FD article"*

Step 2: build the skill yourself

I want a reusable skill (SKILL.md) that turns an academic paper, my own or someone else's, into a news article in the style of the Financial Times or the FD.

First ask me up to three questions to tailor it, one at a time. Then write the SKILL.md:

1. Header: a `name` *and a one-line* `description`*.*
2. Output rules: short messages; one question per message, in plain text; a suggested answer I can accept with "ok".
3. Interview before drafting: am I an author; the one finding; why it matters now (with a news search); who should act; one example; what the paper does not show; quotes. Then three angles to choose from.
4. Drafting rules: open with the most surprising finding, caveat early.
5. Fact check: every number to a page, every claim against the paper's own sentence, every quote to its source. . . .

Sonnet 5.5 · effort: high ∨

Click the box to run this prompt in Claude, or write your own. Full text: `materials/prompt_skill.md`

Step 3: save the skill to your Claude account

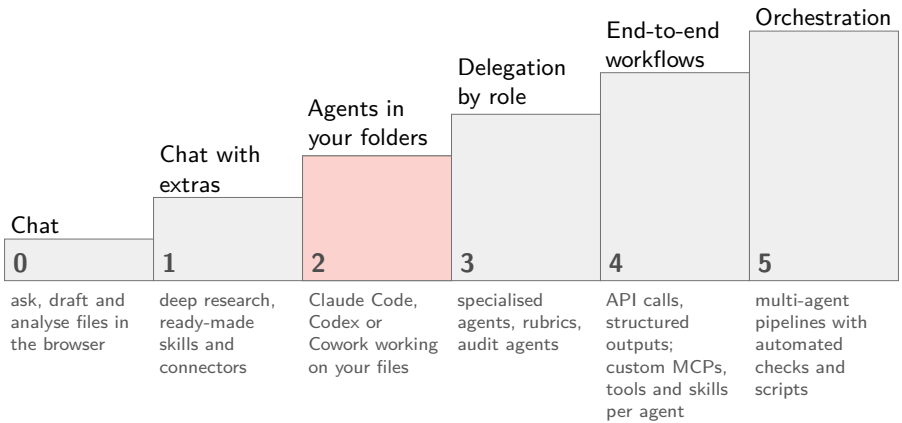
1. In the same chat, ask Claude to package it:

   > *Package this skill as a ZIP file I can upload to Claude: one folder named after the skill, with SKILL.md inside.*

2. Download the ZIP. Do not unzip it
3. In Claude, go to Customize → Skills → + → Create skill → Upload a skill, and choose the ZIP
4. Test it: open a new chat, attach a paper and ask for an FD or FT article. Claude should pick up your skill by itself

Needs *Code execution and file creation* switched on (Settings → Capabilities). Menu names as in Claude's help centre, October 2026.

Let the agents loose: Claude Code, Codex, Cowork



- The agent works in a folder on your own computer: it opens, creates and edits files, fetches data, and writes and runs scripts
- You give it a task and check the result; your skills come along

What's next?

What we have now:

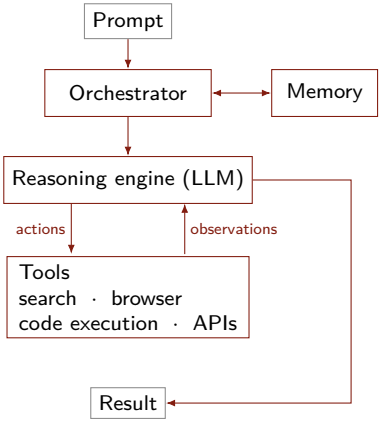
- a decently written article, grounded in the paper
- a link to recent news, and a standard format
- a skill that does this again for the next paper

But a good journalist keeps asking:

- What do I actually know about supply chains?
- The paper is about Hungary. What about other regions?
- Are US banks concerned about the supply chain risk of their borrowers?

To answer that, we need data, and an AI that can work with it in our own folder.

What is an AI agent?



- An LLM with planning, memory and tools, pursuing a goal through multi-step actions
- Orchestrator: ordinary software that runs the loop
- LLM: decides the next step; asks for a tool when it needs one
- Tools: search, browse, run code; the result comes back as an observation
- Loop until done: reason, act, observe

Korinek (2025), "AI Agents for Economic Research", §3.1 and Figure 1; the reason–act–observe loop is the ReAct pattern of Yao et al. (2023).

The agent can now read, edit and create files

Claude desktop app: Cowork

1. Open the Claude desktop app and choose **Cowork**
2. Share a folder with Claude, e.g. `fd-demo/`
3. Describe the task. Claude reads the folder, shows a plan, and saves its work back there

Claude can only read and write inside the folders you share.

Claude Code

1. In the desktop app: **Code** tab → **Local** → select the folder
2. Pick a mode: **Ask** (approve every change), **Auto** or **Plan**
3. Or in a terminal: `cd fd-demo`, then `claude`

Full access to the project, its files and the terminal; asks before running commands. Pro plan and up.

Claude, "Navigating the Claude desktop app: Chat, Claude Cowork, Claude Code", claude.com/resources, read 21 October 2026. Menu names may change: click through once before the session.

Getting data in: download a file, or call an API

Download

The agent finds the dataset on a website and saves the file (CSV, Excel).

Good for a one-off: one table you need once.

"Download the table from CBS StatLine and save it in analysis/data/."

API

An API is a web address that returns data to a program instead of a page to a person. The agent writes a short script that asks for exactly the series it needs.

Good when you want to rerun it: new data, same script.

No key needed: SEC EDGAR (company filings; your email in the request), Eurostat, CBS, ECB. Free key: FRED. Keep keys out of the chat.



every number in the draft can be traced back along this line

Step 4: let an agent work in your folder

Claude Code · folder: `fd-demo/`

In this folder you will find paper.pdf and draft.md.

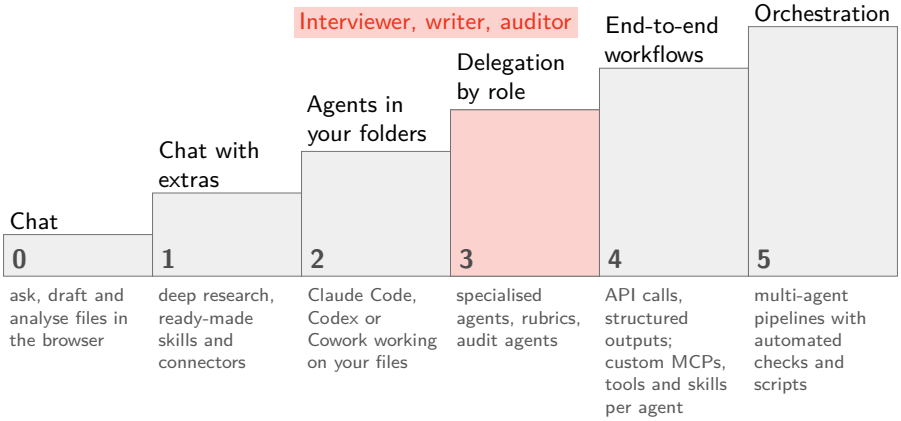
1. Read both.
2. Using the SEC EDGAR API (no key; send a User-Agent header with my name and email; stay under 10 requests per second), download the most recent 10-K of the ten largest US banks: JPMorgan Chase, Bank of America, Citigroup, Wells Fargo, Goldman Sachs, Morgan Stanley, U.S. Bancorp, PNC, Truist and Capital One.
3. In each 10-K, find every sentence that mentions "supply chain". Classify each one: (a) the bank's own operations or vendors, (b) its borrowers' or clients' credit risk, or (c) the general economy.
4. Save the sentences with their bank, filing date and class in a CSV, plus the script and one bar chart (mentions per bank, by class), in a new folder called analysis/.
5. Make your work visible and readable in a notebook as well.

Ask me before installing anything. Stop after step 5.



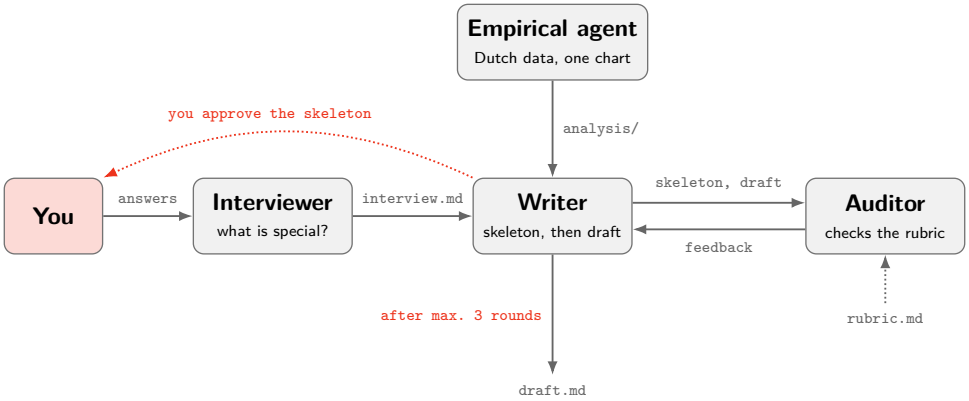
Click the box to open it in your browser, copy it into Claude Code, or type your own. Full text: `materials/prompt_agent.md`

One more level up: agents with roles



For a 600-word article this is overkill: let's do it anyways

Step 5: a team of agents, each with one job



- Each agent gets its own instructions and starts fresh; they pass work through files
- The auditor did not write the text, so it judges it more strictly

How to build a team

```
# .claude/agents/auditor.md
---
name: auditor
description: Checks a draft article against
rubric.md and the paper. Use after every new
draft.
tools: Read, Grep, Glob
model: sonnet
---

You are a strict newspaper editor. You did
not write this draft.

Read draft.md, paper.pdf and rubric.md. Trace
every number to a page. Compare every claim
with the paper's own sentence.

Return at most five problems, worst first. Do
not edit any file.
```

- One text file per agent, in `.claude/agents/` in your project folder. It ends in `.md`, like a skill
- The header between the dashes is YAML: a simple format for settings, one `key: value` per line. It says who the agent is, when to use it, and which tools and model it gets
- Below the header: plain instructions, like a skill
- No `Edit` or `Write` in `tools`: the auditor can read the draft but not change it
- Make one: ask Claude Code to write the file

Claude Code documentation, "Subagents", code.claude.com/docs/en/sub-agents, read 21 October 2026. Works in Claude Code (terminal or the Code tab of the desktop app).

A second agent, and how the team works together

```
# .claude/agents/writer.md
---
name: writer
description: Writes and revises draft.md from
interview.md and analysis/. Use after the
interview and after each audit.
tools: Read, Write, Edit
skills: fd-article
model: opus
---

You write for the Financial Times, following
the fd-article skill.

First write a skeleton to skeleton.md, one
line per paragraph, and stop.

After approval, write draft.md. When you get
the auditor's feedback, fix only those points.
```

Unlike the auditor, the writer may write and edit files, and it loads your skill.

The main Claude Code chat is the manager. It talks to you, calls each agent in turn and passes the results on.

One prompt runs the team:

> *Interview me with the fd-article skill and save my answers in interview.md. Then ask the writer agent for a skeleton and wait for my approval. Then let the writer draft and the auditor check, back and forth, at most three rounds.*

- The interview stays in the main chat: agents cannot ask you questions while they work
- Files are the hand-over: `interview.md` → `skeleton.md` → `draft.md`
- Save this prompt as a skill, and the whole team runs with one line

Claude Code documentation, "Subagents", code.claude.com/docs/en/sub-agents, read 21 October 2026: agents run in sequence through the main conversation; subagents cannot use the ask-the-user tool.

Wrap-up

- One prompt already gives a decent article, but not one we can publish: no voice, nothing checked, no reason to care *now*
- Each level up added something
  - Chat with a skill: your method, written down once: interview first, fact check after
  - Agent in your folder: data and scripts, saved next to your draft
  - A team of agents: separate roles, and an auditor that did not write the text
- Climb only as high as the task needs: for one article, a skill is usually enough
- Whatever the level, you check every number and quote, and could defend every sentence